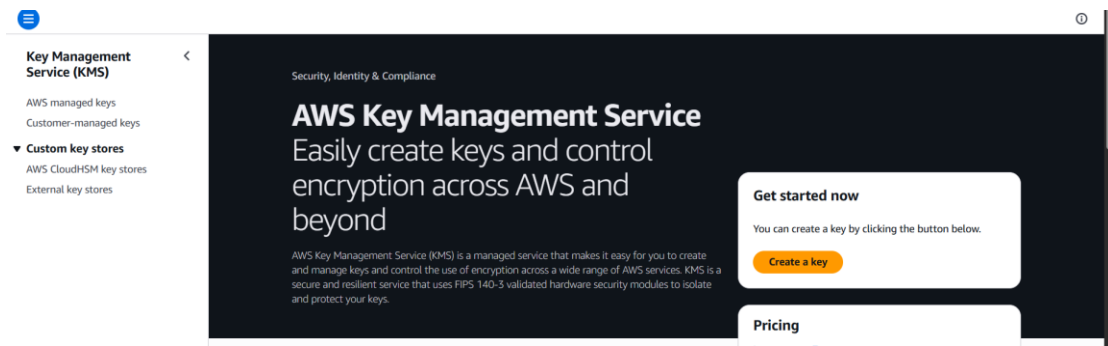


AWS KMS

AWS Key Management Service (KMS) is a managed service that makes it easy to create and control cryptographic keys used to protect your data. KMS integrates with most AWS services — S3, RDS, EBS, Lambda, Secrets Manager and more — to encrypt data at rest automatically.

Implementation Steps

1. Create a new KWS



2. Configure symmetric and encrypt and decrypt

This screenshot shows the 'Configure key' step in the AWS KMS console. A progress bar on the left indicates the current step is 'Step 1: Configure key', with subsequent steps like 'Add labels', 'Define key administrative permissions', etc., shown as optional. The main form has two sections: 'Key type' and 'Key usage'. Under 'Key type', 'Symmetric' is selected, with a description: 'A single key used for encrypting and decrypting data or generating and verifying HMAC codes.' Under 'Key usage', 'Encrypt and decrypt' is selected, with a description: 'Use the key only to encrypt and decrypt data.'

3. Given new name and description

This screenshot shows the 'Add labels' step in the AWS KMS console. The progress bar on the left shows 'Step 2: Add labels' as the current step. The main form has two sections: 'Alias' and 'Description'. Under 'Alias', there is a text input field containing 'my-ecommerce-kws'. Under 'Description', there is a text input field containing 'KMS key for encrypting ecommerce app data'.

4. Given IAM user name and select one of it

Step 3 - optional

Define key administrative permissions

Step 4 - optional

Define key usage permissions

Step 5 - optional

Edit key policy

Step 6

Select the IAM users and roles authorised to manage this key via the KMS API. These administrators will be added to the key policy under the statement 'ic administration of the key'. Modifying this Sid might impact the console's ability to update the administrator statement in the key policy. [Learn more](#)

☐	Name	Path	Type
☐	AdminUser	/	User
☒	Devops	/	User
☐	test-user	/	User

5. Key policy

Configure key

Step 2

Add labels

Step 3 - optional

Define key administrative permissions

Step 4 - optional

Define key usage permissions

Step 5 - optional

Edit key policy

Step 6

Review

Edit key policy - optional

Key policy

Review the key policy statements for this key. To manually update this policy, select **Edit**. Modifying the statement identifier the console displays updates to that statement.

```
1 {
2   "Id": "key-consolepolicy-3",
3   "Version": "2012-10-17",
4   "Statement": [
5     {
6       "Sid": "Enable IAM User Permissions",
7       "Effect": "Allow",
8       "Principal": {
9         "AWS": "arn:aws:iam::928430096503:root"
10      },
11       "Action": "kms:*",
12       "Resource": "*"
13     },
14     {
15       "Sid": "Allow access for Key Administrators",
16       "Effect": "Allow",
17       "Principal": {
18         "AWS": "arn:aws:iam::928430096503:user/Devops"
19       }
20     }
21   ]
22 }
```

6. Key status enabled

Key Management Service (KMS)

AWS managed keys

Customer-managed keys

Custom key stores

AWS CloudHSM key stores

External key stores

Success

Your AWS KMS key was created with alias my-e-commerce-kws and key ID a9c4086e-a0af-4870-8f85-e710ca169d30.

View key

Customer-managed keys (1)

☐

Aliases

☐

my-e-commerce-kws

☐

Key ID

☐

a9c4086e-a0af-4870-8f85-e...

☐

Status

☐

Enabled

☐

Key type

☐

Symmetric

☐

Key spec

☐

SYMMETRIC_DEFAULT